Speaker notes: my slides

Brando Mattivi · architecture + data (4–7) and Module B / price (13–17)

How to use these notes. For each slide you get: the **goal** (what the committee should take away), a spoken script (**Say**), the key **numbers** to land cleanly, and the **transition** to the next slide. Do not read tables and chips row by row: group them and point with your hand.

04 One pipeline, five stages

ARCHITECTURE

Goal: give the mental map of the whole system. Everything the others present later hangs off this slide.

SAY

> The system is one pipeline with five stages. We start from external data (ENTSO-E, weather), which flows through an imputer that repairs the gaps, and produces a clean dataset split into train, validation and test. From there two forecasters branch off: Module A predicts the **load**, Module B predicts the **price**. The key idea is the arrow: each stage does not pass a single number to the next, but **three quantiles, q10, q50 and q90**, that is a band of uncertainty. Finally Module C, a reinforcement-learning agent, consumes that price band to decide how to operate a battery and generate profit.

Close like this (the thesis through-line): two forecasters and one decision agent, each consuming the previous stage's quantiles. Uncertainty is **propagated**, not thrown away.

05 Divider · Part 01 · Data & Imputation

TRANSITION

Goal: a transition slide. *5–10 seconds, do not read it word for word.*

SAY

> We move into the first part: data and imputation. We have six external sources, all with missing data, and before any model sees them we repair them with a self-supervised deep imputer.

→ Move straight to slide 6.

06 Six external data sources

DATA · SOURCES

Goal: show the data is real, heterogeneous and non-trivial to collect. Group it, do not read the table.

SAY

> The six sources fall into three families. From **ENTSO-E**, the European transparency platform, we take four signals: the day-ahead price, the actual load, generation across 17 fuel types, and cross-border flows with France, Austria and Switzerland. Then **weather** from Open-Meteo for five German cities, and finally **fuel prices**, TTF gas and EU-ETS carbon, which are the fundamental drivers of the electricity price.

In total about `61,000 hourly rows`, from `2019 to 2025`, which become `52 columns`.

DETAIL THAT LANDS (IF YOU HAVE TIME / IF ASKED)

> One non-obvious technical point: ENTSO-E uses **two different EIC codes**, one for the market zone and one for the physical zone. Market and physical do not coincide, and using the wrong code simply returns empty data.

Transition: but the real problem is not the amount of data, it is the quality.

07 Missing data is structured, not random

DATA · QUALITY

Goal: the key conceptual slide of the data block. Message: the gaps are not random, and that changes how you treat them. Slow down here.

SAY

> When we looked at the missing data, we found it is not random, it is **structured**, and that distinction is crucial. There are three cases. First: **coal-gas** has no records on ENTSO-E at all from 2019 to 2022, the data does not exist, it is not a gap to fill. Second: **nuclear dropping to zero** on 15 April 2023 is the real shutdown of the German plants, a true value, not a missing one. Third: scattered gaps from holidays or exchange closures, which are short and recoverable.

THE CONCLUSION (THE POINT)

> A naive fill, like a mean or forward-fill across everything, would inject **fake signal** where there is no information. Long structural gaps must be left alone; only the short ones should be imputed. And that is exactly what the method on the next slide does.

Transition (to the imputation colleague): from here I hand over to [name], who explains how the imputer automatically distinguishes these cases.

13 Divider · Part 03 · Price forecasting

TRANSITION

Goal: a short transition slide, but it is "yours": you can lean into it a little.

SAY

> We reach the heart of the project, price forecasting. The idea is per-quantile gradient boosting, with conformally calibrated intervals carrying a finite-sample guarantee. I will walk you through it in the next slides.

14 Per-quantile boosting, then conformal

MODULE B · ARCHITECTURE

Goal: explain the architecture. Two messages: three independent boosters, and boosting is a justified choice, not a default.

SAY

> We start from **33 causal features**: each one uses only past information, never the future, which is essential to avoid cheating through leakage. They sit in six families: calendar, price lags, the spark and dark spreads (the gas and coal margins), spike flags, a 2022-regime indicator, and weather. Then the key point: instead of one model predicting all quantiles together, we train **three independent CatBoost models**, one for q10, one for q50, one for q90. Finally the conformal calibration, **CQR**, which adjusts the band.

THE CHOICE TO OWN (BOTTOM LINE)

> We also tried a deep LSTM, but on this data, which is repetitive and categorical-heavy, **gradient boosting won**. It is a choice driven by results, not a default.

Transition: but the raw boosters had a calibration problem, and I will show it honestly.

15 We fixed our own miscalibration

MODULE B · CALIBRATION

Goal: tell a mistake of yours and how you diagnosed and fixed it. Committees love this. On the left "before" (red dots pierce the band), on the right "after".

SAY

> At first our boosters were **over-confident**: the band was too narrow and the true values pierced it constantly. You see it on the left, the red dots above the band. Concretely: 35% of the true values sat above q90, when by definition it should be only 10%. The fix is **conformal quantile regression**. It measures on the validation set how much we miss and widens the band by exactly that quantile of the error, with the finite-sample formula $\lceil (n+1)(1-\alpha) \rceil / n$, which gives a **theoretical coverage guarantee**, not just an empirical one.

Result: the upper-tail miss drops from `35.5% to 6.8%`, very close to the ideal 10%.

PRE-EMPT A QUESTION

> A deliberate choice: we target an **80% band, q10–q90, not 95%**. The extreme tails are rare, unpredictable spikes; 80% is the robust interval that is actually useful for the downstream decision.

Transition: calibrated this way, let us see how accurate it is.

16 46% better than seasonal-naive

MODULE B · RESULTS

Goal: the headline results. Let it land, this is the numbers slide.

SAY

> These are the results on the **locked test set, the first quarter of 2025**, data the model never saw. The metric is MAE in euros per megawatt-hour, lower is better. The two baselines: the naive at 24 hours sits at 47.6, the seasonal-naive at one week at 43.3. Our production model, **CatBoost plus CQR, drops to 23.3**: a **46% improvement over the seasonal-naive**.

Chips to cite: coverage `0.82` (the band covers 82% of cases, in line with the 80% target), and the **Diebold-Mariano test with** $p < 0.001$: the gap from the baseline is statistically significant, not luck.

Honest note (optional): LightGBM was close too, 24.8. CatBoost wins by a small margin: both confirm boosting is the right choice here.

Transition: I close with a result we expected to be positive, and instead it was not.

17 Load did not help price

MODULE B · HONEST NEGATIVE

Goal: the honest negative. It earns scientific credibility: a null result reported, not hidden.

SAY

> A natural hypothesis was that Module A's load forecasts would help predict the price. We wired them into Module B as extra features, measured, and the result was an improvement **close to zero**: no material gain on the price MAE.

THE METHODOLOGICAL POINT (THE REAL MESSAGE)

> We report it instead of burying it. In serious work, null results must be stated.

THE CLOSING THAT KEEPS THE DOOR OPEN

> This does not mean load is useless. It is useless only here, for predicting the price. Load **does matter, but later, at the dispatch stage**, where the agent decides how much to charge and discharge the battery. And my colleagues will show exactly that in Module C.

Final transition (to Module C): with this I close the price part and hand over to [name] for the battery dispatch.

Quick delivery tips

- **The three numbers to land perfectly:** 46% better than baseline · MAE 23.3 €/MWh · drop 35.5% → 6.8% on calibration.
- **Do not read tables and chips row by row.** Group them and point with your hand.
- **Slides 5 and 13 are transitions:** 5–10 seconds, no more.
- **Slides 7, 15 and 17 are your method slides:** that is where you show you understood the *why*, not just the *what*. Slow down on those.
- **Likely committee questions:** why boosting and not deep learning · why 80% and not 95% · what exactly the finite-sample CQR guarantees · why load does not help price.